



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name :

Team Leader Name :

Problem Statement :

Team Members

Team Leader:

Name:

College:

Team Member-1:

Name:

College:

Team Member-2:

Name:

College:

Team Member-3:

Name:

College:

Opportunity — the Zero-Isolation mission

“No Indian community should be digitally invisible — before, during, or after a disaster.”

- **Problem:** floods & landslides sever roads and bridges and silently isolate hospitals. In the 2015 Chennai floods, 100+ patients were cut off for 48 hours (evacuated by fishing boats); 14 died at MIOT Hospital when isolation killed power and oxygen.
- **How we are different:** we fuse optical imagery with ISRO's NISAR radar to see roads through cloud and canopy, quantify uncertainty, and turn roads into a resilience digital twin — most teams stop at a road mask.
- **Our USP:** the Zero-Isolation Engine + Time-to-Isolation (TTI) — it predicts which single failure will isolate a hospital, and the minimum-cost fix, before disaster strikes.
- **Research contribution:** a confidence-aware resilience graph that fuses satellite-derived infrastructure health, uncertainty-aware road extraction, and network-resilience simulation into one operational decision model.

Features — an operational resilience capability

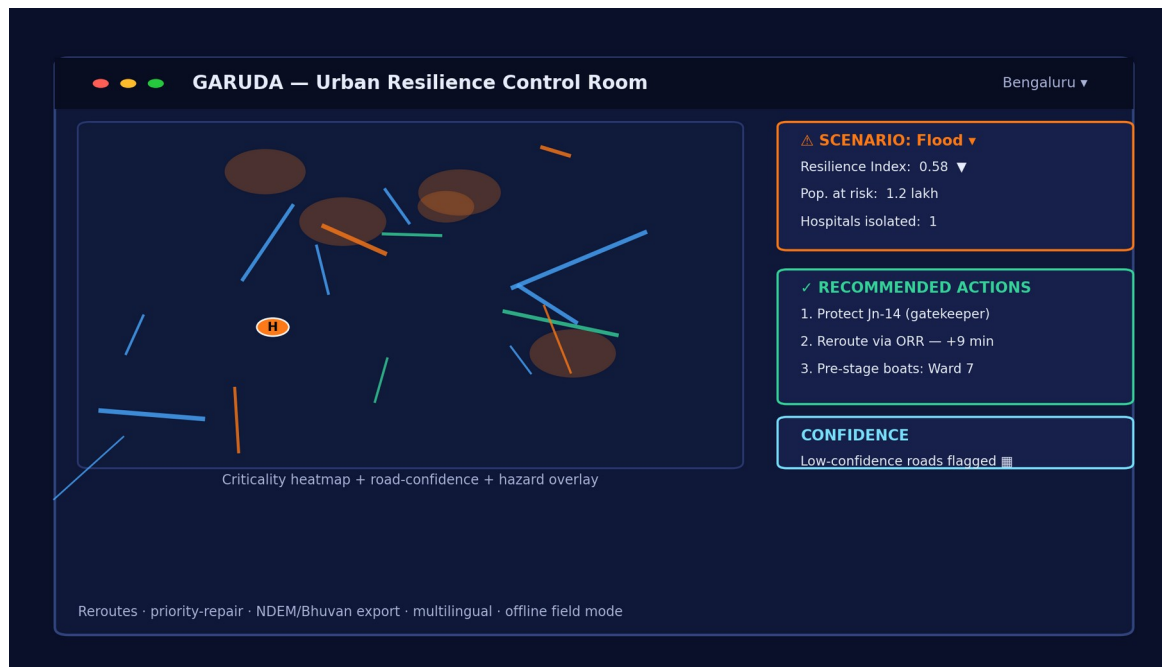
- **Zero-Isolation Engine** — finds every single point of failure that isolates a hospital, and the minimum-cost fix.
- **Time-to-Isolation (TTI)** — hours until each hospital / school / fire-station is cut off.
- **City Pulse** — a live city resilience score (0-100).
- **Guardian Mode** — always-on; the “Next Failure” panel warns before officials ask.
- **Uncertainty-aware extraction** — flags low-confidence roads; never hallucinates.
- **Infrastructure Memory** — temporal change from repeated EO (deformation, subsidence): uniquely satellite.
- **Recovery Intelligence** — post-disaster damage, repair priority & reconstruction monitoring (full lifecycle).
- **Investment optimization** — “₹50 cr → strengthen these 4 bridges → +18% city resilience.”
- **Explainability** — “why critical? 47% of hospital traffic; alternative +23 min; 93% confidence.”
- **Plus** — pre-monsoon audit, pre-positioning advisor, NDEM/Bhuvan export.

Process flow — a closed learning loop, not a pipeline



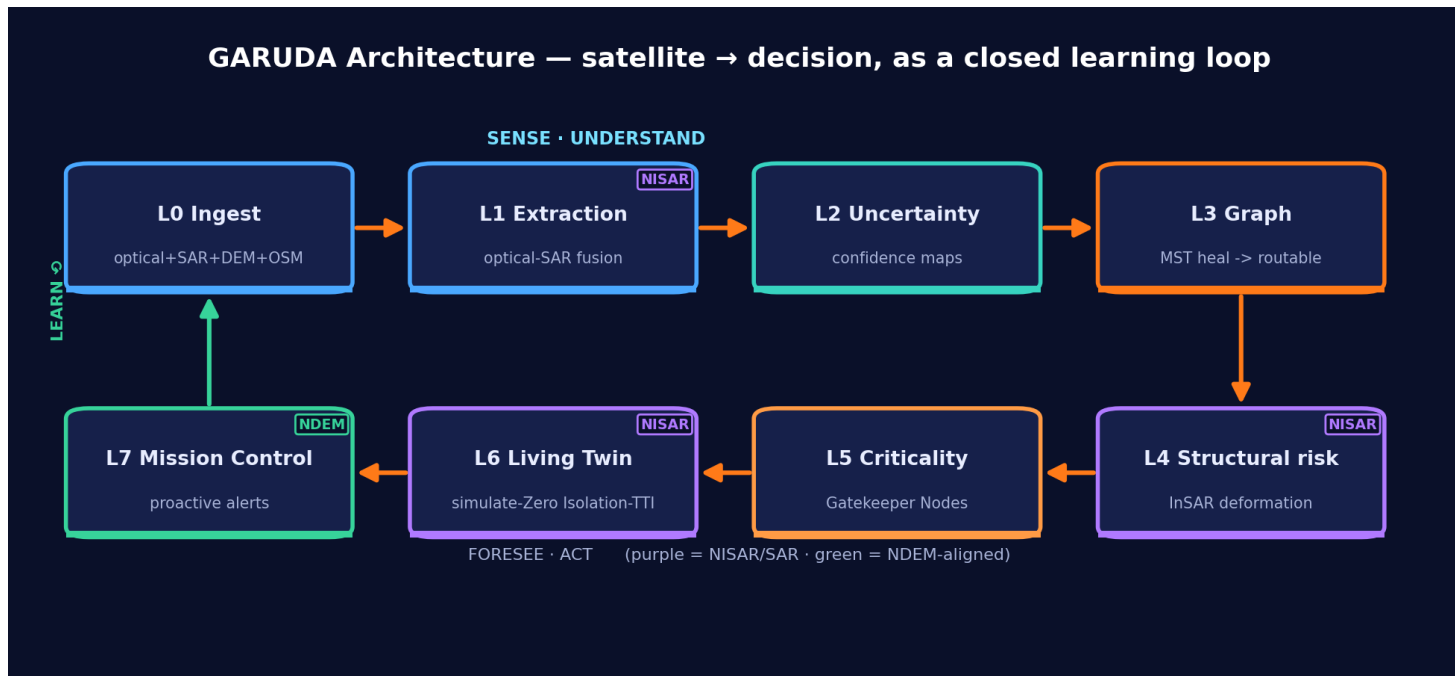
Hero flow: rain → NISAR flags a weak flyover → the twin simulates the flood → City Hospital isolates ~150k people → Guardian Mode alerts 36 h early → one bridge fix keeps it connected.

Wireframe — GARUDA Mission Control



Live map (criticality · confidence · hazard) · the “Next Failure” panel (predicted failure, confidence, TTI, population, hospitals, action) · City Pulse · recommended actions · NDEM/Bhuvan export.

Architecture — L0 → L7, satellite to decision



Optical-SAR fusion → uncertainty → routable graph → NISAR-InSAR structural risk → criticality → Living Twin (Zero-Isolation, TTI) → Mission Control. Purple = NISAR/SAR · green = NDEM. PoC = one pilot city; multi-infrastructure & live ingestion = roadmap.

Technologies

- **ML** — PyTorch, SegFormer / Mask2Former (ViT), Albumentations; uncertainty via MC-dropout / evidential DL.
- **Geospatial** — GDAL, Rasterio, ESA SNAP (SAR), GeoPandas, OSMnx, scikit-image.
- **Graph & simulation** — NetworkX (betweenness, global efficiency, percolation).
- **Product** — FastAPI, Streamlit / React + Leaflet/Mapbox, PostGIS.
- **Data** — NISAR & Sentinel-1 (SAR); Sentinel-2 / LISS-IV / Cartosat-3 (optical); CartoDEM, OSM, IMD; SpaceNet/DeepGlobe — via Bhoonidhi & Copernicus.
- **Compute** — Google Colab / Kaggle (free GPU).

Estimated implementation cost — frugal & indigenous

- **Data — ₹0:** open & indigenous (Bhoonidhi, Copernicus, OSM, IMD); Cartosat-3 provided at the finale.
- **Compute — ₹0:** Google Colab / Kaggle free GPU for the PoC; modest cloud GPU only at national scale.
- **Software — ₹0:** a fully open-source stack.
- **People:** the 3-4-member team + ISRO mentorship.
- **Bottom line:** a near-zero-cost proof of concept — aligned with ISRO's Atmanirbhar, budget-conscious mandate.



HACK2SKILL

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THANK YOU